

7. INTRODUCTION INTEGRATED CIRCUITS

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Integrated Circuits play a very important part in electronics. Most are specially made for a specific task and contain up to thousands of transistors, diodes and resistors. Special purposes IC's such as audio-amplifiers, FM radios, logic blocks, regulators and even a whole micro computers in the form of a micro controller can be fitted inside a tiny package. Some of the simple Integrated Circuits are shown in figure 7.1.

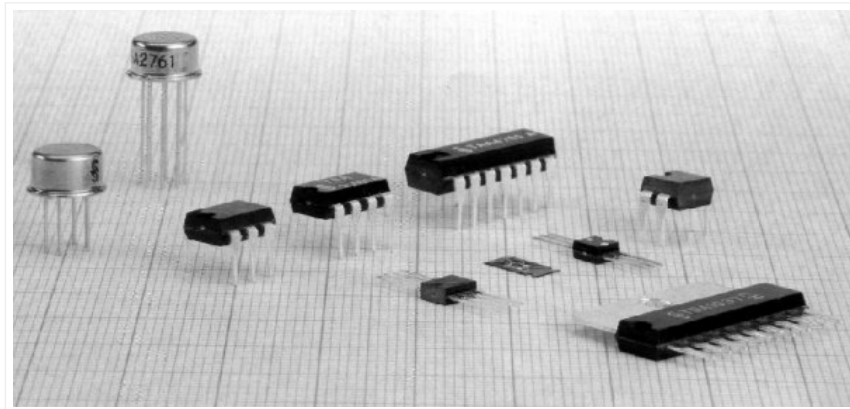


Fig. 7.1: Integrated circuits

Depending on the way they are manufactured, integrated circuits can be divided into two groups: hybrid and monolithic. Hybrid circuits have been around longer. If a transistor is opened, the crystal inside is very small. This means a transistor doesn't take up very much space and many of them can be fitted into a single Integrated Circuit.

The pin-out for some of the common packages is shown in Figure 7.2:

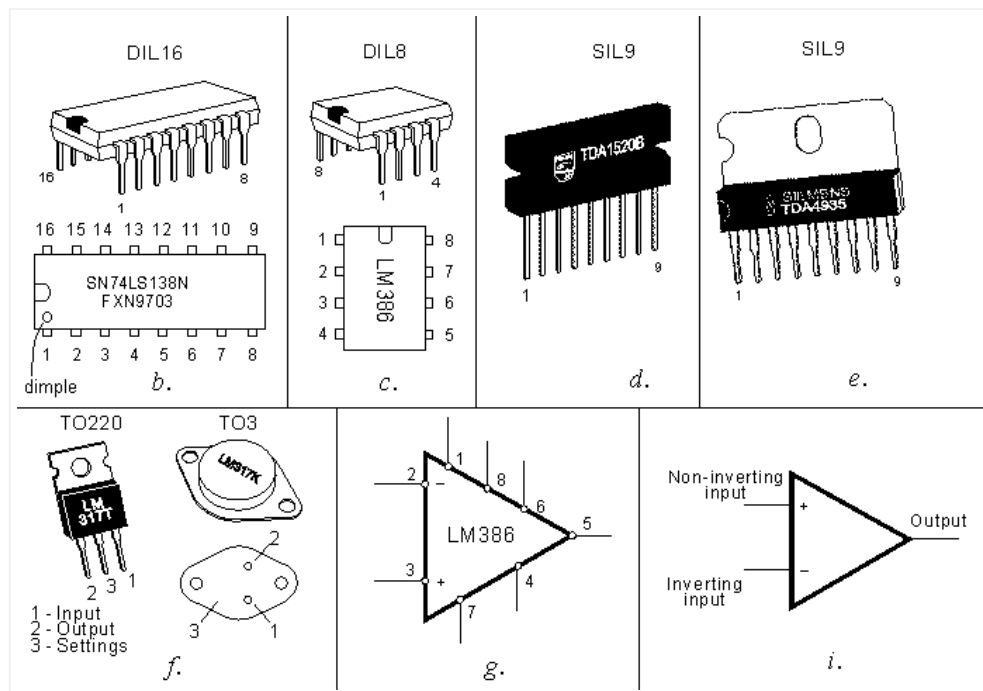


Fig. 7.2: Pin-out and symbols for some common integrated circuits

Most integrated Circuits are in a DIL package – Dual In Line, meaning there are two rows of pins. (DIL16 and DIL8 are shown in 7.2b and 7.2c). The device is viewed from the top and the pins are numbered in an anti-clockwise direction.

High power integrated circuits can generate a lot of heat and they have a metal tag that can be connected to a heatsink to dissipate the heat. Examples of these IC's are shown in 7.2d and 7.2e, and 7.2f.

Symbols used to represent integrated circuits are shown in 7.2g and 7.2i. Symbol 7.2g is commonly used to represent amplifiers.

Figure 7.2i shows an operational amplifier. Signs + and – represent inverting and non-inverting inputs. The signal to be amplified is applied between one of the inputs and ground (ground and supply aren't represented, but are necessary for the circuit to operate).

Integrated circuits can be divided into two further groups: **analog (linear) and digital**. The output voltage of a linear circuits is continuous, and follows changes in the input. Typical representative of a linear IC is an integrated audio amplifier. When a signal from a microphone is connected to the input the output will vary in the same way as the voltage from the microphone. If watched on an oscilloscope, the signal on the output will be the same shape as the mic's signal, only the voltage will be higher depending on the amplification of the integrated circuit. It is a different situation with digital IC's. Their output voltage is not continuous. It is either LOW or HIGH and it changes from one state to the other very quickly.



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